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Report on Pricing Pure Pet Insurance Premiums

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Table of Contents

Executive Summary

This report outlines the process to develop an interpretable model for predicting pure premiums for pet insurance. It is known that different claim conditions vary in frequency and severity, with chronic conditions like allergies generally having lower severity and higher frequency compared to acute conditions (Morrell, 2014). As such, we applied k-means clustering to group risks by condition categories with similar claim size, volatility, and frequency. We also incorporated qualitative intuition as it is unlikely that clear risk groups have fully formed after just 1 year of claims data (Kaas, 2008).

Our combined quantitative and qualitative approach suggested the use of three risk levels to strike an optimal balance between performance and interpretability. Broadly, we observed that chronic health issues typically presented lower risk, while acute health events, though less frequent, often resulted in higher claim severities.

Leveraging the identified clusters, we experimented with various machine learning algorithms—ranging from GLM and shrinkage methods to tree-based and advanced techniques—to predict claim frequency and severity. Inspired by Caruana et al. (2019), we proposed an innovative model - *Explainable Boosting Models (EBM)*, which extend Generalised Additive Models (GAMs) by using boosted trees to capture feature effects independently. This combines the interpretability of GLMs with the predictive accuracy of boosting, addressing GLM limitations such as the inability to capture variable interactions automatically (Lou, 2012). To account for the limited claims data, which might not satisfy the data requirements of a semi-parametric model, we applied a credibility adjustment. This allowed initial predictions to incorporate results from the broader severity model, with greater reliance on cluster models as data volume grows. The final model was an EBM after credibility with a RMSE of 701.56, an 8.16% improvement over a baseline GLM.

The condition category modelling and use of EBMs delivered several insights for specific claim categories with high confidence due to the robust model performance. For instance, pets with higher expected aggression tend to incur more behavioural claims, especially those with a long-life expectancy. We also observed global trends, such as the influence of socioeconomic factors on severity and the significance of pet age, excess, and life expectancy across all models.

Methodology

First, the earned dataset underwent initial cleaning to remove extraneous information from several date columns. Both the `UW_date` and `nb_policy_first_inception_date` included a default time-of-day standard, which was deemed unnecessary. These were initially parsed as character data and were subsequently converted to the correct date format. Additionally, inconsistent labels in `pet_de_sexed_age` (e.g., '0-3 mo' vs. '0-3 months') were standardised, and empty values were replaced with 'Never de-sexed' for clarity. The cleaned data was then aggregated to ensure each `exposure_id` was represented by a single row, with `earned_units` reflecting the total exposure period. There were 12 exposures with NA values for their date of birth, and these were subsequently removed.

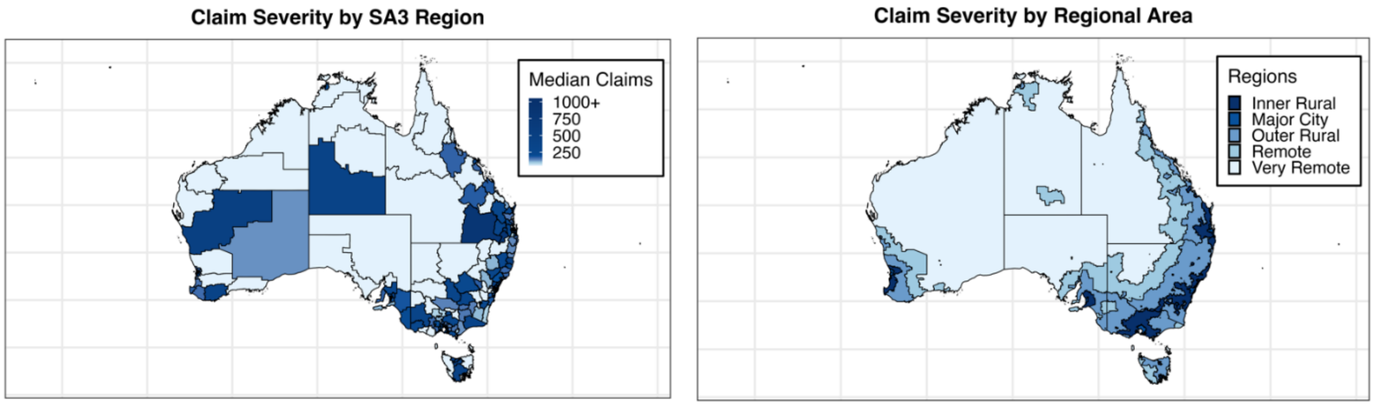
The claims experience data was imported and cleaned, removing all rows where `claim_paid` was zero. Rather than merging frequency and severity data, we modelled individual claim sizes, following an approach suggested by Frees (2014) to minimise information loss. Policyholder information was integrated into the claims data, which was then used to calculate claim counts by condition category.

The data was then divided into training (80%) and test sets (20%) using stratified sampling in the validation set approach. The training set was used for model development with 5-fold cross-validation for parameter tuning, while the test set was reserved for evaluating performance. The Root Mean Square Error (RMSE) and Gini Index were chosen as performance metrics, providing measures of prediction and ranked distribution accuracy, ensuring robust performance.

Exploratory Data Analysis (EDA)

Incorporating Postcode Information

The initial model development focused on feature engineering, using Heaton's (2016) approach to explore interactions and nonlinearities through EDA. To incorporate postcode information, we used ABS income, economy, and community data to approximate policyholder socioeconomic status. SA4 regions proved too broad and SA2 too detailed with missing values. Ultimately, SA3 regions (Figure 1(a)) provided an ideal balance of detail and generalisation, while regional groupings (Figure 1(b)) captured broader geographic trends, consistent with Laflamme's (2002) findings on urban-rural claim variability. This provided a two-pronged approach for capturing claim variability across postcodes.

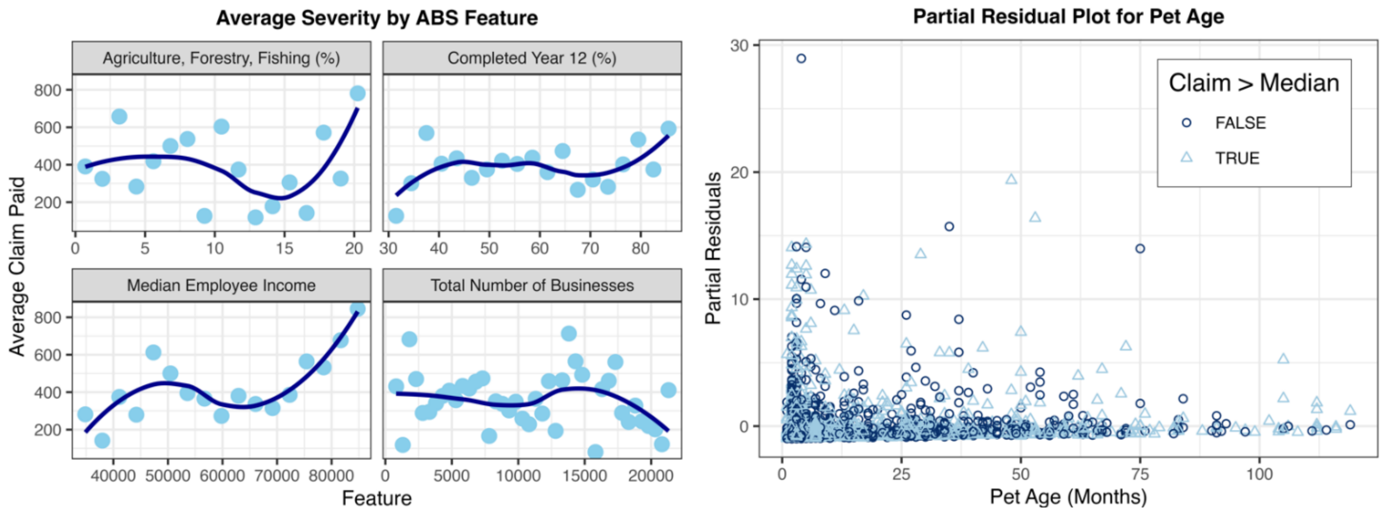


(a) Median Claims by SA3 Region (b) Median Claims by Region

Figure 1: Incorporating Postcode Information

Non-linear Relationships

Figure 2(a) depicts several non-linear relationships between numeric variables and claim severity (appendix for frequency). Notably, regions with higher percentages in Agriculture, Forestry and Fishing exhibited increased severity, likely due to rural lifestyle risks like work-related injuries (Myers, 2011). Additionally, claim severity rises in areas with incomes above $60,000, possibly due to wealthier individuals having improved access to veterinary care (Goldman, 2002). The partial residuals for pet age from a baseline GLM further indicate that this model fails to capture non-linearities in the data (Figure 2(b)). Accounting for this, we elected to explore non-linear models in model construction.

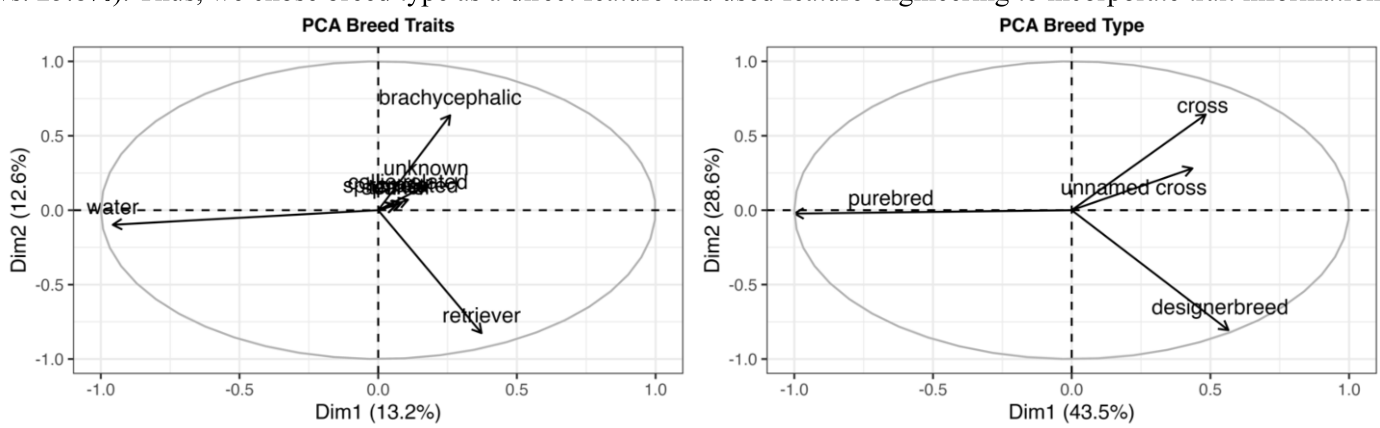


(a) Median Claims by SA3 Region (b) Baseline GLM Partial Residuals

Figure 2: Evidence of Non-linear Relationships

Principal Components Analysis (PCA)

Figure 3(a) reveals high correlations among breed traits, suggesting redundant information. Conversely, Figure 3(b) demonstrates clearer distinctions between breed types, with the first two components explaining more variance (72.1% vs. 25.8%). Thus, we chose breed type as a direct feature and used feature engineering to incorporate trait information.

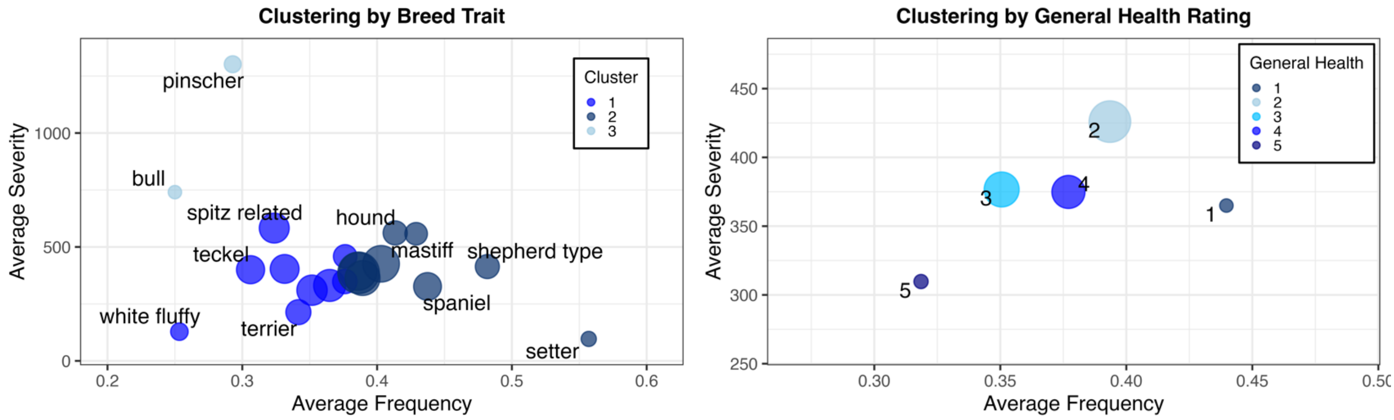


(a) Breed Traits (b) Breed Type

Figure 3: Principal Components Analysis (PCA)

Breed Characteristics

K-means clustering was used to examine specific breeds impact claim costs, a common trend in pet insurance (O'Neill, 2014). However, Figure 4(a) shows many breeds with similar claim frequencies and severities, likely due to the portfolio's young age where breed-specific claim patterns emerge later with chronic issues (Hayward, 2016). Hence, we incorporated breed-specific behavioural traits from the 'Dog Breeds' dataset, a compilation of data from newsletter 'DogTime'. This took the form of ordinal information regarding the adaptability, sociability, health, trainability, and exercise requirements, of different breeds. Ultimately, providing clearer distinctions (Figure 4(b)) and quicker insights, enabling a proactive approach to claim cost predictions which overcomes the delayed onset of breed-specific claims.



(a) Breed Traits (b) `General.Health` from new dataset

Figure 4: 2-D Representation of k-means clustering (see appendix for 3-D representation)

Feature Engineering Summary

To maximise the relevancy of the new datasets, we performed further feature engineering to link the new data to the pre-existing policyholder information, an approach Berrada (2022) states will yield more accurate risk assessments. The first added variable, `exp_future_life`, took the difference between the breed life expectancy from the Dog Breeds dataset and the insured animal's age, providing an intuitive measure of life expectancy for each dog currently in the portfolio. Next, we drew on findings from Zink (2014), showing reduced aggression in desexed dogs, and Duffy (2008), which examined breed-specific aggression. This informed the creation of `expected_aggression`, serving to predict the dog's sociability. Here, a zero indicated that the animal was desexed, while non-desexed animals were assigned numbers 1 through 5, each with diminishing friendliness value. We also added the `dog_safety` variable, which combines `is_multi_pet_plan` status with general breed friendliness. A value of 0 indicates a dog without other pets, while higher values reflect decreasing friendliness toward other dogs, serving as a predictor for dog-on-dog injury incidents. Finally, `happy_in_apartment` was constructed as a combination of policyholder address type and the breed's ability to adapt to apartment life as given in the Dog Breeds dataset. Here, 0 indicated the animal was not living in an apartment, while apartment-dwelling animals were rated from 1 to 5 based on their breed's suitability for apartment living. This variable replaced the address type, offering a more tailored measure of how lifestyle influences the specific animal.

Clustering

To begin model construction, we grouped claims by similar condition categories, a method shown to improve risk estimates in pet insurance (Johnson, 2021). The rationale for this was our hypothesis that a clustered approach would reveal which factors influence specific claim types, offering more nuanced insights into risk assessment. We used k-means clustering to classify condition categories by similar risk levels based on claim size, volatility, and frequency. The elbow plot (see appendix) suggested two possible cluster counts, with $k = 3$ achieving a silhouette score of 0.445 and entropy of 0.860, and $k = 5$ scoring 0.367 and 1.468, respectively. Although the quantitative results favoured three clusters, Kaas (2008) advises blending quantitative performance with domain knowledge, as risk groups may be less distinct with limited data. Thus, we opted for the five clusters suggested from k-means, and then consolidated four categories into one as shown by the enclosed region in Figure 5. As shown in Table 1, the combined approach resulted in a slight reduction in performance compared to the quantitative approach. However, this was offset by improved interpretability and more consistent data distribution across clusters (Table 2), which will enhance modelling reliability.

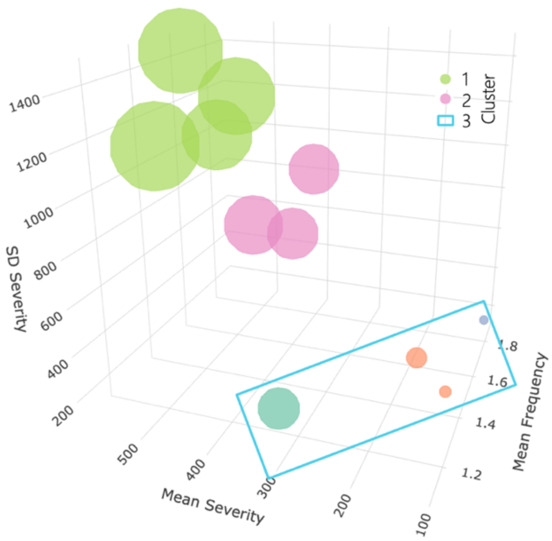


Figure 5: Final Clusters

K=3	Silhouette Score	Entropy
Quantitative Approach	0.445	0.860
Combined Approach	0.362	0.923

Table 1: Comparison of Clustering Performance

Number of Observations	n1	n2	n3
Quantitative Approach	2279	766	37
Combined Approach	1702	577	803

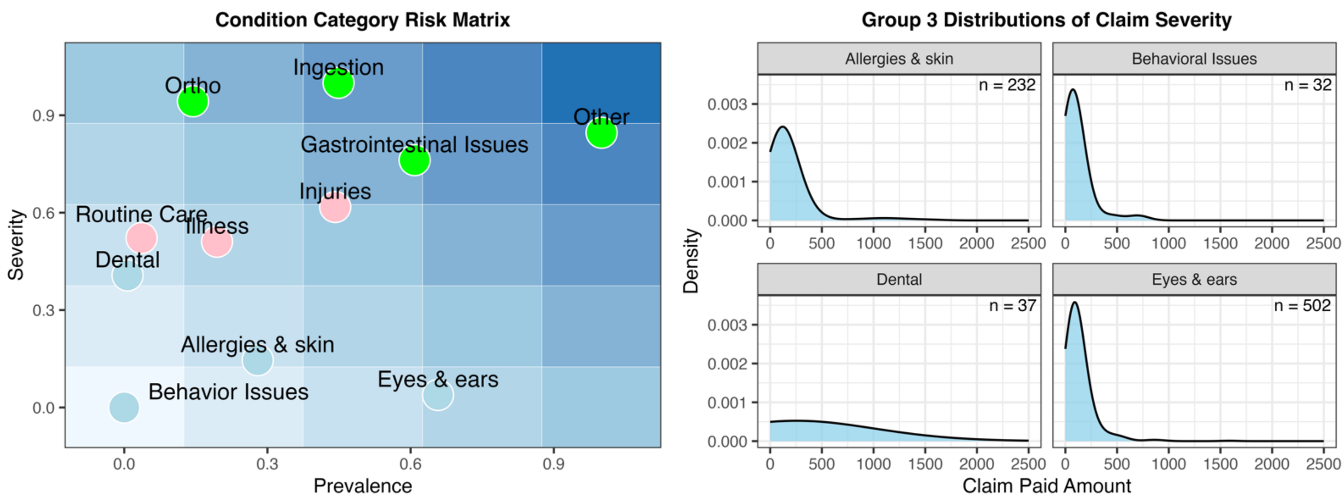
Table 2: Data for Severity Modelling

Clustering Insights and Recommendations

Figure 6(a) summarises the key results in a risk matrix, incorporating 'prevalence' to give the business a clearer context of the claim types they will regularly encounter. Cluster 1 features in the top right corner are especially concerning for their high prevalence and severity, including claims from chronic conditions like orthopaedic, gastrointestinal, ingestion, and miscellaneous issues. These cases have high expected severity and variability, as their acute nature forces pet owners into costly, urgent decisions (Clark, 2020). Anderson (2019) highlights that these claims often involve complex surgeries, with age and weight significantly impacting costs, an insight supported by modelling results (Figure 11(a)).

Cluster 2 claims were less common but carried a moderately high severity risk, encompassing both preventative and reactive care, including injuries, illnesses, and routine care. Unlike Cluster 1, which involves chronic, worsening conditions, Cluster 2 consists of isolated, unpredictable issues without underlying concerns. The high severity of routine care may stem from customers combining routine visit costs with additional vet-recommended procedures (Volk, 2022). To address this from a practical standpoint, regular collaboration with vets is advised to ensure transparent billing.

Cluster 3 was classified as lower risk by the matrix, encompassing claims related to chronic conditions requiring ongoing management, including allergies & skin, eyes & ears, behavioural issues, and dental care. This group shows low average severity and low variability, representing chronic health issues that require ongoing management rather than sudden, costly surgeries like in Cluster 1. As shown in Figure 6(b), dental claims had a flatter distribution with a heavy right tail. Clustering was difficult due to limited data ($n = 37$) and the dual nature of claims: routine maintenance resulting in small costs and occasional expensive surgeries like root canals. Harvey (2005) emphasises that dental issues in older dogs are common and expensive, suggesting that as the portfolio matures, severity will increase. Hence a revised claims process is recommended to differentiate routine care from surgical procedures and better manage risk.



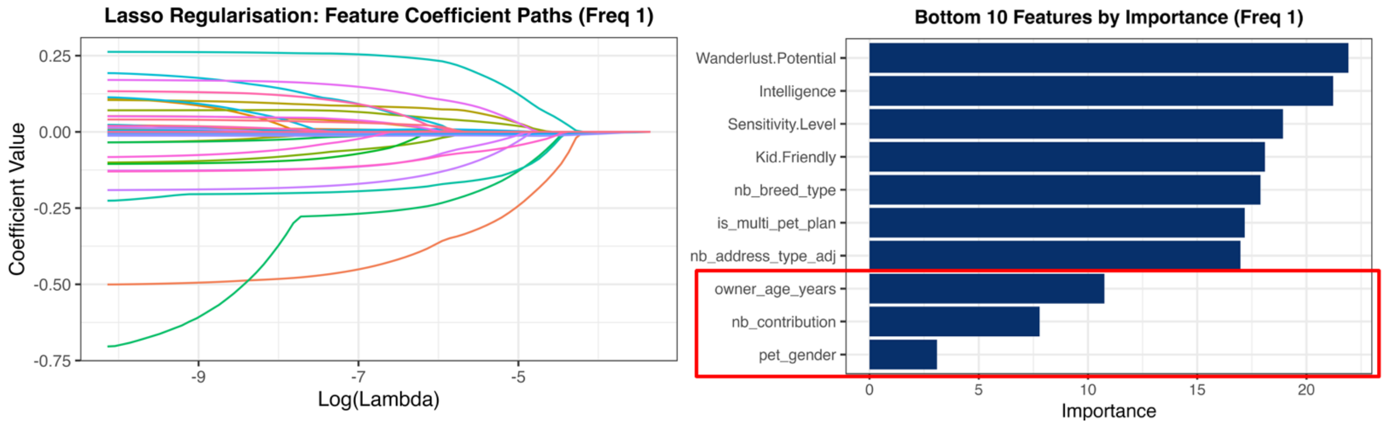
(a) Median Claims by SA3 Region

(b) Baseline GLM Partial Residuals

Figure 6: Insights Gained from Clustering

Construction of the Claims Model

To remain competitive in the pet insurance market, the pricing model must balance accuracy and innovation. It also needs to be highly interpretable to guide product design, meet regulatory requirements, and clearly convey risk factors. With these objectives in mind, various machine learning methods were evaluated, with detailed analysis in the appendix. Within each method, six models were developed based on the identified clusters, using a blended approach of lasso regularisation and random forests to select the most important features before modelling. For instance, features such as `owner_age_years`, `nb_contribution`, and `pet_gender` demonstrated low importance and were quickly regularised to 0 in Frequency Model 1 for the EBM, leading to their removal (Figure 7). Parameters were tuned to minimise RMSE using 5-fold cross-validation, while the Gini Index was also considered to select a robust model across several metrics.



(a) Feature Coefficient Paths (b) Random Forest Feature Importance

Figure 7: Example of Output Used to Remove Features from Models

GLM Methods

Generalised Additive Models was the best performing GLM method with a lower RMSE (721.04) and higher Gini Index (0.481), exceeding the baseline GLM's RMSE of 763.88 and Gini Index of 0.240. This approach used 3 Poisson GLMs for frequency and 3 Gamma GLMs for severity, with splines applied to numeric features. A zero-inflated Poisson model was also developed to account for the high proportion of policyholders with no claims (Figure 16). However, its performance was mixed, improving in RMSE (760.51) and Gini (0.251) over the GLM, but still trailing the GAM.

Shrinkage Methods

Both shrinkage methods showed comparable performance. Lasso achieved a lower RMSE (740.38) and higher Gini Index (0.316) than the GLM, while Ridge slightly improved RMSE (739.45) but had a marginally lower Gini Index (0.315) than Lasso. With the GAM as the top GLM performer, applying Lasso regularisation further improved it, reducing RMSE to 719.96 and raising the Gini Index to 0.522, making it the best performing model thus far.

Tree-Based Methods

Boosting outperformed other tree-based methods with a RMSE of 702.70 and a Gini Index of 0.571, surpassing the best shrinkage model (GAM with Lasso). In contrast, the regression tree was highly interpretable but had poor metrics, with a RMSE of 765.98 and a Gini Index of 0.238. Random forests and bagging were also tested, with Random forests achieving a lower RMSE of 708.63, though still falling short of boosting (see Table 5 for detailed comparisons).

Advanced Methods

Since boosting achieved the lowest RMSE of all models, an XGBoost model was then trained. This extends boosting to incorporate regularisation, resulting in a minor reduction in RMSE (701.30). A neural network was also trained, yielding the lowest RMSE of all models (696.44) and highest Gini Index (0.588). One limitation of neural networks and tree-based models is despite their strong performance, they are highly uninterpretable. As such, we decided to research ways to incorporate the predictive power of trees with the high interpretability offered by GLMs. This led us to the Explainable Boosting Model (EBM), introduced by Caruana et al. (2019). The EBM combines the additive structure of GAMs, with the flexibility of boosting. This approach allows the model to maintain the transparency of a GLM while automatically capturing complex interactions typically only accessible to more advanced models. This hypothesis was supported by the modelling results, yielding comparable RMSE (702.46) and Gini Index (0.564) to that of the boosting model. Given that the regularised models outperformed their non-regularised counterparts (e.g., boosting and XGBoost), we enhanced our approach by combining Lasso and random forests for feature selection (Figure 7).

Selected Model

The best model for predicting claim frequency and severity was the EBM with Lasso and random forests for feature selection, offering strong performance comparable to boosting while retaining GLM-like interpretability. Optimal parameters were chosen using 5-fold cross-validation with grid search, prioritising simplicity to prevent overfitting. For example, Frequency Model 1 used 500 boosting rounds, a 0.05 learning rate, and a minimum of 20 samples per split.

A major concern with this semi-parametric model was its high data requirements, particularly for severity models in clusters 2 and 3, which had fewer than 1,000 training rows (see Table 7). To address this, we applied the Bühlmann-Straub credibility model (1970), blending cluster-specific severity predictions with a model trained on all data. The credibility factor, based on the cluster's data size and response variance, determined the optimal weighting. For clusters 1 and 3, high credibility factors indicated reliable predictions, while cluster 2's factor of 40.8% suggested using 59.2% of the overall model's predictions. This adjustment further reduced RMSE (701.56) and increased the Gini Index (0.566), enhancing model robustness. The selected model was then re-trained using the entire given dataset and used to generate predictions for the prospective customers.

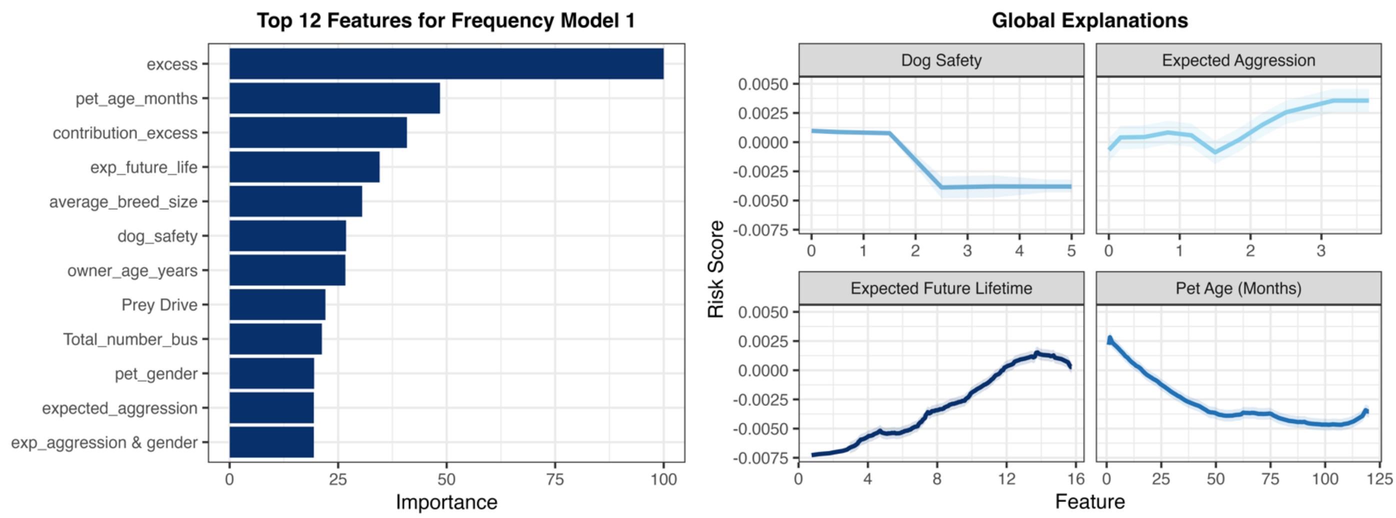
Claims Characteristics

Global Characteristics

The EBM model's high interpretability, with its additive structure, allows 'global explanation' plots to reveal each feature's impact on risk scores. Combined with the modelling approach of clustering by condition category, this allows for nuanced insights into key risk drivers for specific claim types. In all models, `nb_excess`, `pet_age_months`, and `exp_future_life` emerged as key predictors driving risk. As shown in Figure 8(b), claim frequency generally increases with expected lifetime but declines with age, reflecting fewer claim opportunities as lifespan shortens. For severity, pets with an expected future lifetime of $5-8$ years were identified as high risk, as the longer lifespan allows more time for severe health issues to emerge as shown in Figure 9(a) for Severity Model 1.

Factors Impacting Frequency and Severity for Dental, Allergies & Skin, Face, and Behaviour

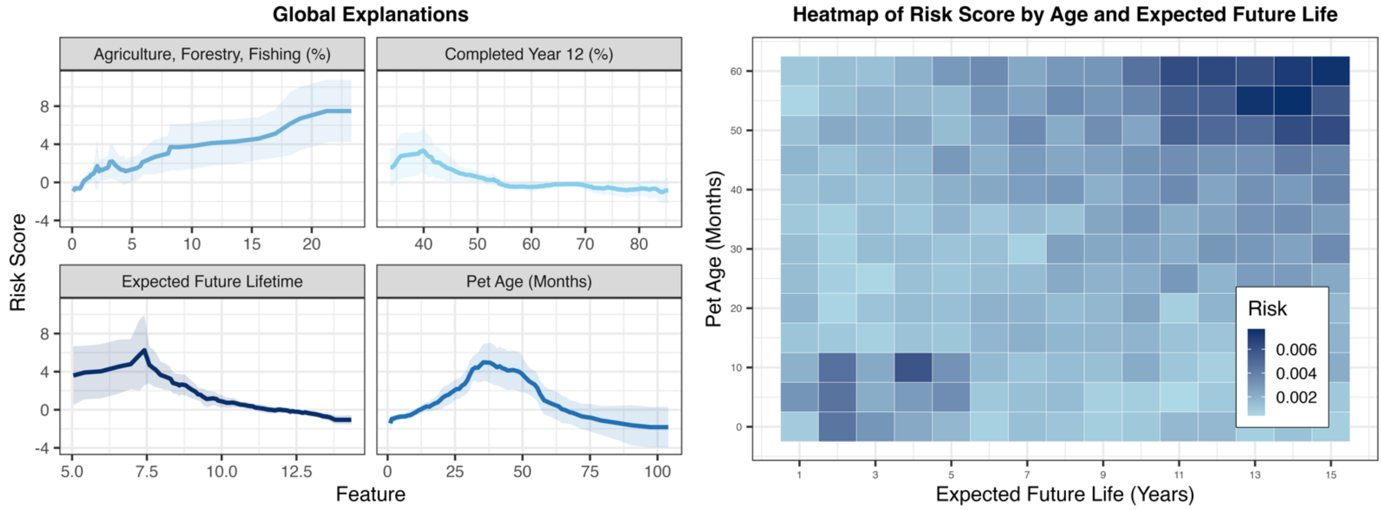
As illustrated in Figure 8(a), `dog_safety` emerged as a highly significant predictor. Figure 8(b) reveals that pets with low ratings $(0, 1, 2)$ increase the risk score, aligning with NCOIL (2021) findings that multiple pets together can heighten claims for allergies, skin, and eye/ear issues due to increased interaction. This was reinforced by the high importance of `expected_aggression`, shown in Figure 8(a). The line plot indicates that higher expected aggression in dogs correlates with increased claim frequency, supporting ASPCA (2021) suggestion that desexing aggressive breeds may reduce behavioural claims. This insight is deepened by the interaction between `pet_gender` and `expected_aggression` as shown in the importance plot, where male dogs with high expected aggression are predicted to have higher frequency.



(a) Bar plot of Feature Importance (b) Global Explanation Line Graphs

Figure 8: Interpreting Frequency Model 1

Across all severity models, the ABS income data also proved significant. Specifically, regarding Cluster 1 Figure 9(b) shows that as the Agriculture (%) increases in the policyholder's region, claim severity is also likely to increase. This could be attributed to dogs in rural areas being more exposed to outdoor allergens, which can trigger severe skin and eye allergies. Moreover, Figure 9(a) shows that as the percentage of people who have completed year 12 increases, claim severity steadily declines. Sterling (2024) associates this with increased awareness of preventative healthcare practices for pets, where educated owners are likely to invest in routine veterinary visits to prevent more severe cases of dental, and skin related issues. Figure 9(b) highlights a significant interaction identified by the EBM model: older pets with higher expected future lifetime exhibit higher claim severity. This is likely due to dental and joint issues worsening with age, which Anderson (2017) suggests increases healthcare costs. Additionally, pets with low expected future lifetimes but young ages are also flagged as high-risk. This may reflect congenital or early-onset health issues, which can lead to costly treatments and management from an early age (Fleming et al., 2011).

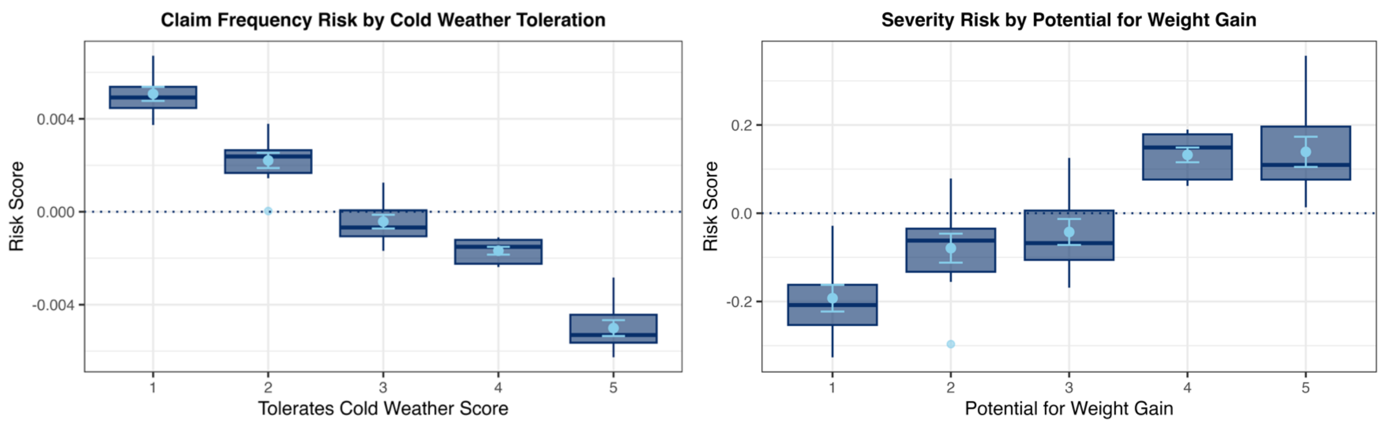


(a) Bar plot of Feature Importance (b) Heatmap of Age and Life Expectancy

Figure 9: Interpreting Severity Model 1

Factors Influencing Frequency and Severity for Illness, Injuries and Routine Care

Figure 10(a) shows that as a dog's tolerance for cold weather improves, the frequency of claims for illness, injuries, and routine care decreases. Buffington (2008) attributes this to cold-tolerant dogs having adaptations like thicker coats or higher body fat, making them less prone to illness. This is of particular concern in Australia, where the volatile climate requires dogs to be highly adaptable. The model also identified a dog's potential for weight gain as a key predictor (Figure 10(b)), with higher weight gain potential linked to increased claim severity. One interpretation is that obesity complicates routine care by increasing anaesthetic risks, raising the cost and complexity of procedures (Laflamme, 2012); furthermore, overweight dogs tend to have weaker immunity and are more prone to injury. For severity, other significant predictors included ABS economic variables, where areas with better economic conditions are more likely to have severe claims. For example, Figure 21 shows a general increasing trend for completed year 12 (%).

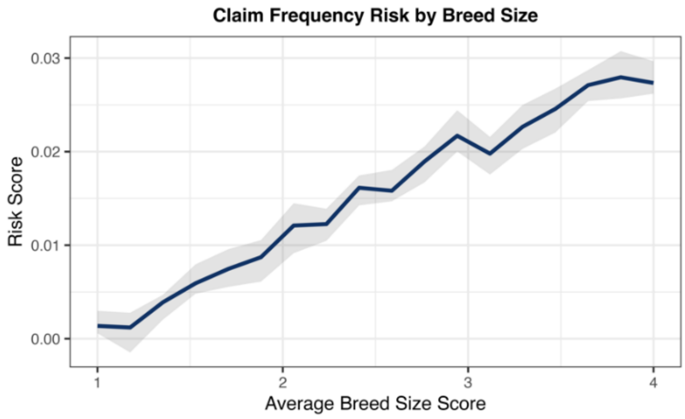


(a) Boxplot of Cold Weather Tolerance (b) Boxplot of Weight Gain Potential

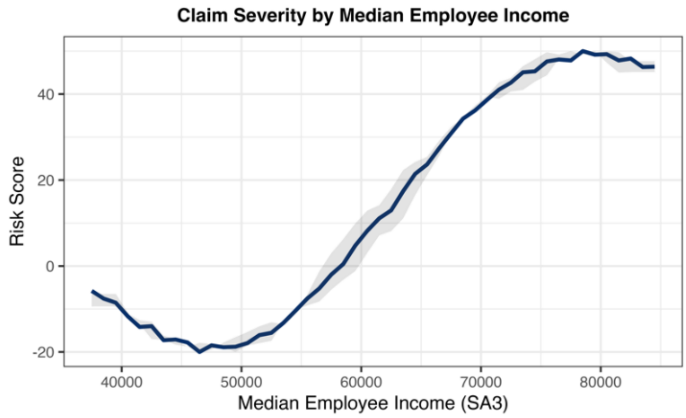
Figure 10: Insights for Group 2 Claims

Factors Influencing Frequency and Severity for Ortho, Other, Gastro and Ingestion

Frequency analysis for Cluster 3 indicates that breed size is a significant predictor for orthopaedic, gastrointestinal, ingestion, and miscellaneous claims (appendix), with claim frequency increasing as breed size grows. Larger dog breeds are more susceptible to orthopaedic issues like arthritis due to the increased stress on their joints, which raises claim frequency as these conditions often require ongoing treatment (Evans, 2010). Moreover, larger breeds tend to have bigger appetites and are more prone to ingesting foreign objects, increasing the frequency of gastrointestinal issues that may require surgical intervention (Buffington, 2008). The breed type and de-sexing status were also highly significant (appendix) as they influence both genetic predispositions to health conditions and behaviours that may lead to injuries or ingestion incidents. Median employee income was a key predictor for claim severity, showing an upward trend for incomes above $50,000. In higher-income areas, pet owners are more likely to afford and choose advanced treatments, leading to increased claim severity; for example, procedures like joint replacements are both accessible and more commonly pursued by wealthier owners (Lund et al., 1999). Following similar logic, the total number of businesses and participation rate were also important features (appendix), suggesting that in more active economic areas, individuals are more likely to have larger claims, reflecting greater access to advanced treatments.



(a) Line Graph of Breed Size



(b) Line Graph of Median Income

Figure 11: Insights for Group 3 Claims

Limitations

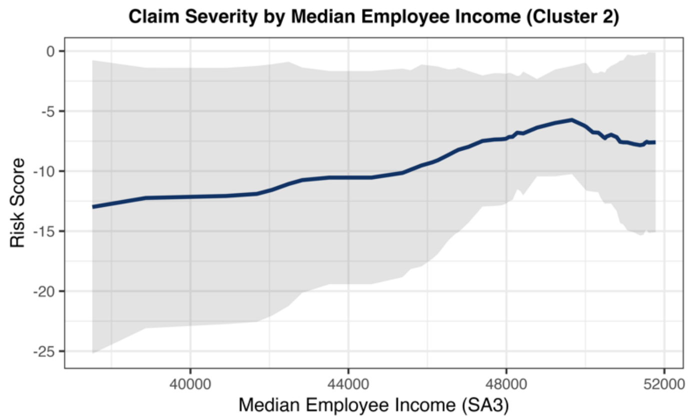
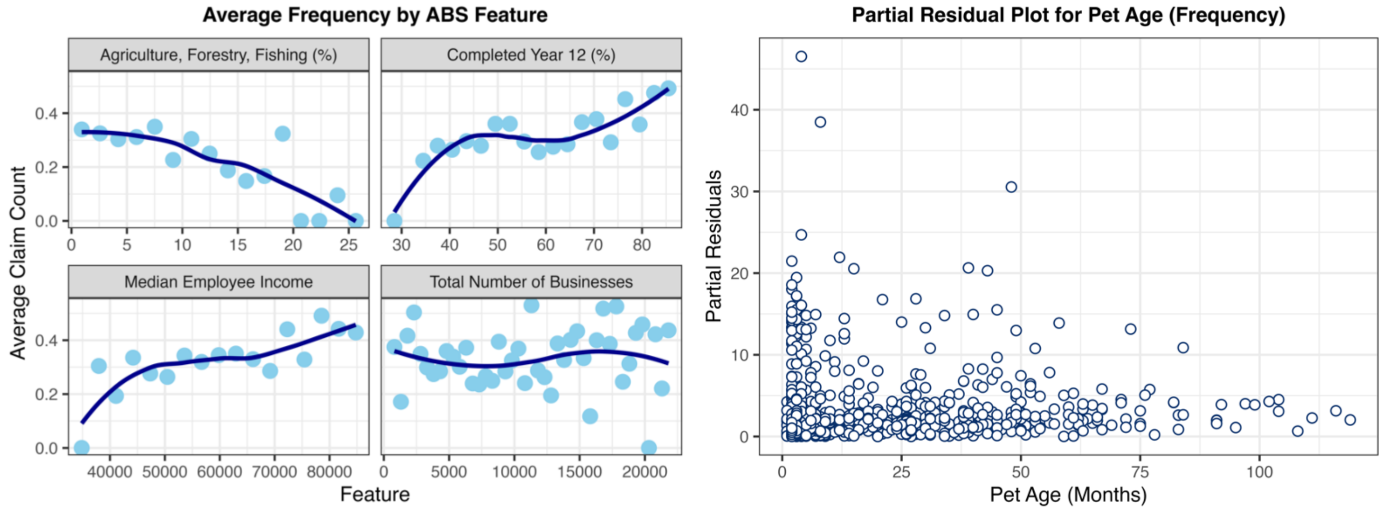


Figure 12: Claim Severity by Income (Cluster 2)

The modelling results were influenced by several limitations inherent in the data sources. In particular, limited data for certain claim categories created challenges in predictive modelling, as semi- and non-parametric machine learning models like EBMs generally require large datasets for reliable performance. This was particularly pronounced in claim severity for cluster 2, where there was only 441 rows of training data; the effect of this can be seen in Figure 12, where insufficient amounts of data resulted in large confidence intervals for predictions. This limitation was partly addressed by the credibility adjustment in severity modelling; however, this reduced the model's interpretability and efficiency. Another limitation was the computational intensity of extensive cross-validation grid searches needed for selecting optimal predictors in the selected EBM model, especially since six separate models were required. Additionally, the EBM lacks a mechanism to prevent it from predicting negative values. Consequently, adjustments were necessary to ensure non-negative predictions in frequency modelling. Alternative models, like zero-inflated Poisson, could have better suited frequency modelling, while EBMs remained effective for severity. Moreover, potential regulatory concerns exist due to relatively new nature of EBMs. Regulators may not be familiar with this model, making it uncertain whether future regulatory changes could impact its use.

Appendix

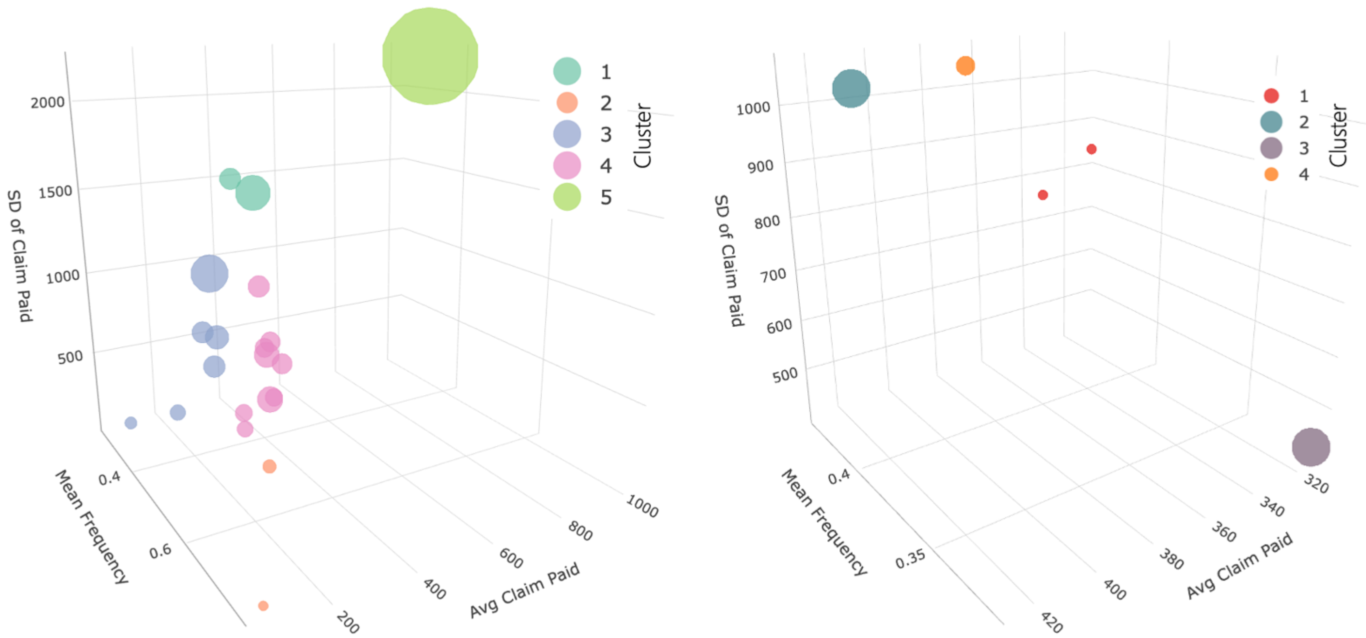
Non-Linear Relationships - Frequency



(a) Claim Frequency by SA3 Region (b) Baseline GLM Partial Residuals

Figure 13: Evidence of Non-linear Relationships for Frequency

Additional Clustering Outputs



(a) Clustering by Breed Trait (b) Clustering by `General.Health` Rating

Figure 14: 3-D Clustering Representation

K-means Clustering Elbow Plot

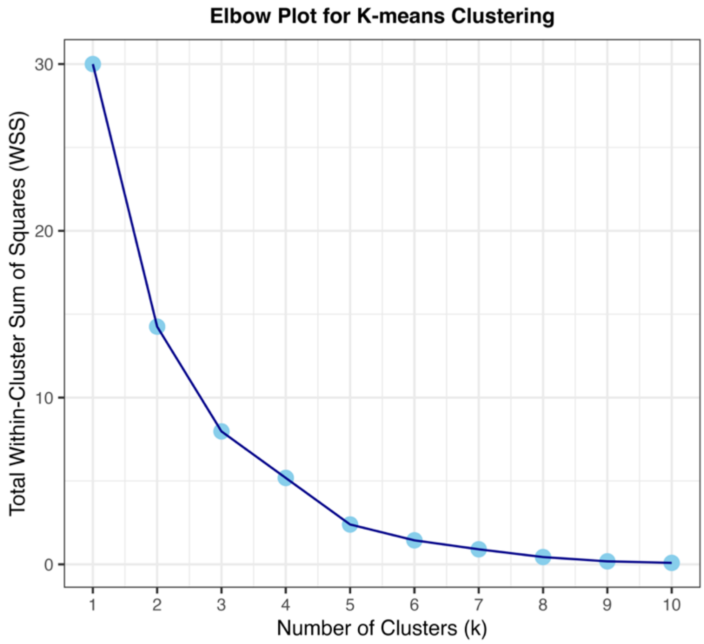


Figure 15: Elbow plot showing two possible elbows at $k = 3$ and $k = 5$

Modelling Results and Analysis for Model Construction

Generalised Linear Model (GLM)

The baseline GLM used a Poisson distribution for frequency and a Gamma distribution for severity, both with a log link. Frequency models included an offset of the log of earned_units, while severity models did not require an offset as they modeled individual claim severities. This model underperformed compared to more advanced methods, with a high RMSE of 763.88 and a low Gini Index of 0.240, providing a baseline for performance comparison.

Zero-inflated GLM

The zero-inflated GLM approach replaces the frequency model with a zero-inflated Poisson distribution to handle the excess zeros in the count data, as illustrated in Figure 16. This model effectively captures data heterogeneity, resulting in a slight reduction in RMSE (760.51) and an improved Gini Index (0.251), indicating a modest enhancement over the baseline GLM.

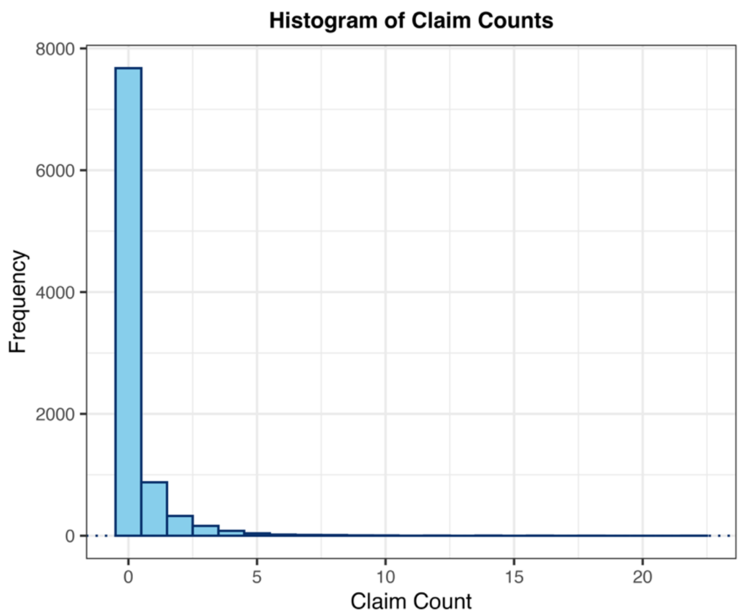


Figure 16: Histogram of Claim Counts

GAM

Generalized Additive Models (GAMs) enhance traditional linear relationships between predictor and response variables by incorporating smooth splines. In this approach, Poisson and Gamma models were applied, using natural splines to address non-linear patterns in numerical features. Optimal degrees of freedom for each feature were determined through 5-fold cross-validation, ensuring a robust balance between flexibility and performance. This strategy significantly improved key metrics, achieving an RMSE of 721.54 and a Gini coefficient of 0.481, indicating that the model's increased adaptability effectively captures complex data patterns without succumbing to overfitting.

Method	RMSE	Gini
Baseline GLM	763.88	0.240
Zero-inflated GLM	760.51	0.251
GAM	721.04	0.481

Table 3: RMSE and Gini Index for GLM Methods

Regression with Lasso

Lasso regression introduces regularization to linear models, applying a penalty to the absolute values of the coefficients to achieve feature selection and shrinkage. By constraining less important coefficients to zero, Lasso effectively simplifies the model, reducing complexity and preventing overfitting. In this analysis, optimal penalty parameters were selected using 5-fold cross-validation, ensuring the best trade-off between bias and variance. This approach resulted in a substantial reduction in RMSE (740.38) and an improved Gini coefficient (0.317), demonstrating Lasso's capability to enhance model performance by focusing on the most predictive features. Whilst significantly improving upon the GLM, the performance was not as strong as the GAM, suggesting non-linear patterns have been missed.

Regression with Ridge

Ridge regression enhances linear models by introducing a regularization term that penalizes the square of the coefficients, reducing multicollinearity and preventing overfitting. Unlike Lasso, Ridge shrinks all coefficients but never reduces them entirely to zero, retaining all predictors while minimizing complexity. This method yielded similar results in RMSE (739.45) and Gini Index (0.317) to the Lasso model.

GAM with Lasso

A Generalized Additive Model (GAM) with Lasso combines the flexibility of smooth splines for capturing non-linear relationships with the regularization power of Lasso. By adding a penalty to the absolute values of the coefficients, this approach simultaneously smooths numeric features and performs feature selection. This led to a large improvement in RMSE (719.96) and a higher Gini Index (0.522), demonstrating the best results of all GLM and regularisation methods.

Method	RMSE	Gini
Lasso	740.38	740.38
Ridge	739.45	0.317
GAM with Lasso	719.96	0.522

Table 4: RMSE and Gini Index for Shrinkage Methods

Regression Tree

Regression trees model are highly interpretable and model the relationship between predictor and response variables by recursively splitting the data into homogenous subsets based on feature values. Optimal splitting criteria and tree depth were determined using 5-fold cross-validation. The final regression tree had slightly worse RMSE (765.98) and Gini Index (0.238) compared to the GLM, suggesting that the increased interpretability is not enough to offset the poor performance.

Random Forest

Random Forest builds an ensemble of decision trees, averaging their predictions to improve accuracy and robustness. By using bootstrap sampling and selecting random subsets of features for each split, Random Forest reduces overfitting and enhances generalisation. Optimal hyperparameters, such as the number of trees and maximum depth, were tuned using 5-fold cross-validation. This approach led to significant improvements in RMSE (708.63) and a Gini coefficient of 0.556, demonstrating Random Forest's strength in capturing complex, non-linear patterns

Boosting

Boosting involves successive models being trained with an emphasis on previous mistakes, with the final result being an ensemble of the trained models This specific iteration used the gbm algorithm. Here, the learning rate and maximum depth of trees were optimised using cross validation with a grid search. This produced RMSE (702.70) and Gini Index (0.571) that were the best out of all models so far, and a marginal improvement over the random forest.

Method	RMSE	Gini
Regression Tree	765.98	0.238
Random Forest	708.63	0.556
Boosting	702.70	0.571

Table 5: RMSE and Gini Index for Tree-based Methods

XGBoost

This specific algorithm of boosting is designed to improve the speed and performance of the modelling boosting process. In addition to the previous boosting parameters, it also introduces a L1 and L2 regularisation parameters, which are tuned also using 5-fold cross validation. This yielded a slight improvement in RMSE (701.30) and Gini Index (0.572).

Neural Network

Neural Network models involve the creation of hidden layers of neurons, which are combinations of the base parameters which these are then used as the basis for predicting the response variable. Due to the use of this model purely as comparison to the others, the learning rate, layers, and neurons parameters were not tuned due to time constraints. Nonetheless, this produced the strongest RMSE (696.44) and Gini Index (0.588) observed of all the models, though these were not significantly above those observed in models such as the XGBoost.

Credibility Adjusted EBM

See Selected Model section of the main report.

Method	RMSE	Gini
XGBoost	701.30	0.572
Neural Network	696.44	0.588
EBM	702.46	0.564
Cred Adjusted EBM	701.56	0.566

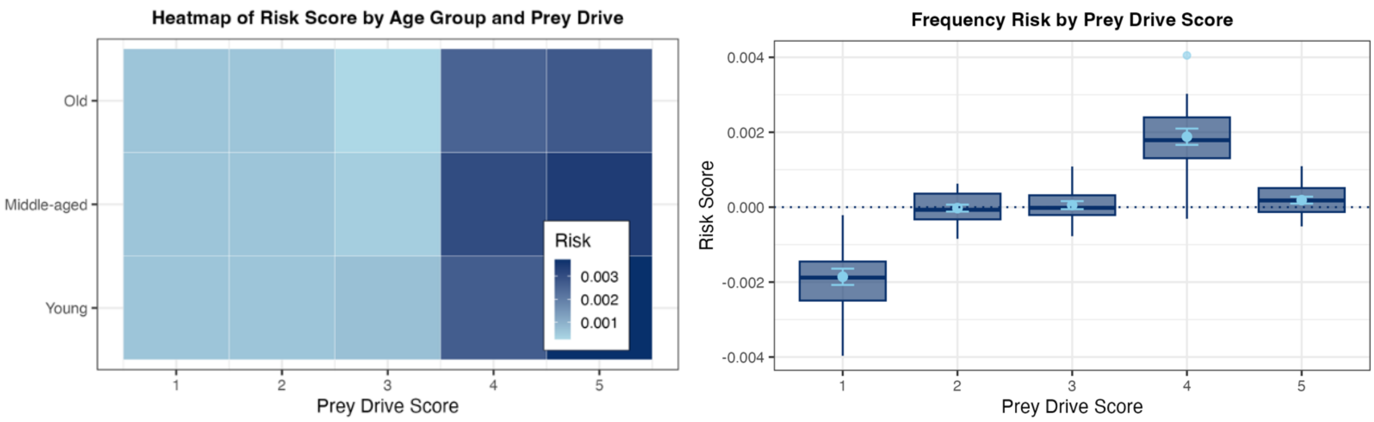
Table 6: RMSE and Gini Index for Advanced Methods

Credibility Key Results

Cluster	1	2	3
Training n	628	**441**	1322
Within Risk Variability (sd)	189.71	**874.13**	1286.14
Credibility Factor (z)	0.999	**0.408**	0.934

Table 7: Credibility Factors for EBM Model

Analysis of Prey Drive for Group 1 Frequency Model



(a) Heatmap (b) Boxplot

Figure 17: Plots to Analyse Prey Drive for Cluster 1 Frequency

Variable Importance Plots for EBM Models (not in main report)

Cluster 1 Models

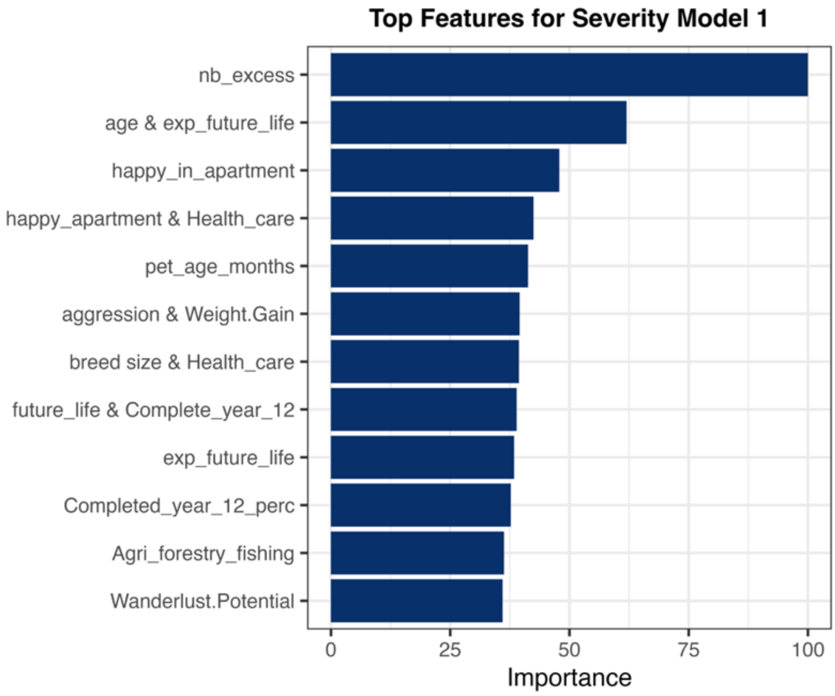
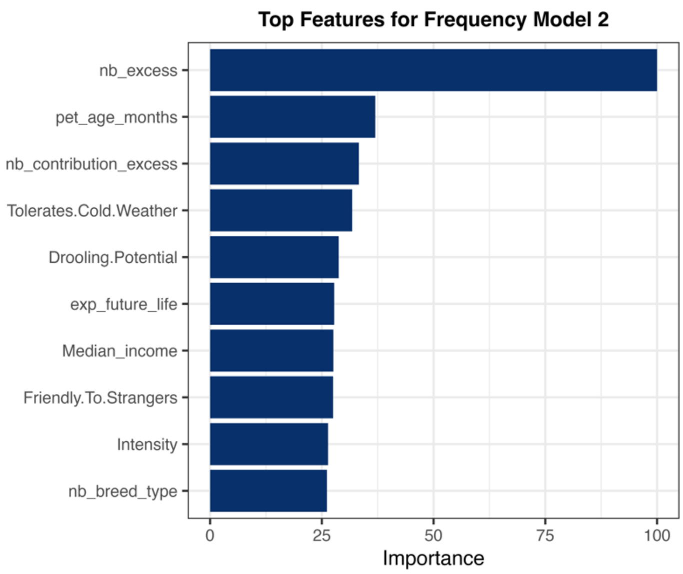
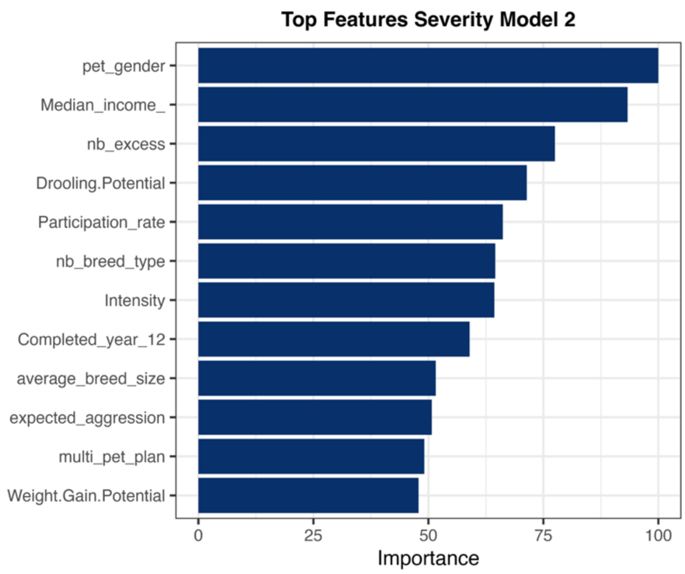


Figure 18: Variable Importance Plot from EBM for Cluster 1 Severity (see main report for frequency)

Cluster 2 Models



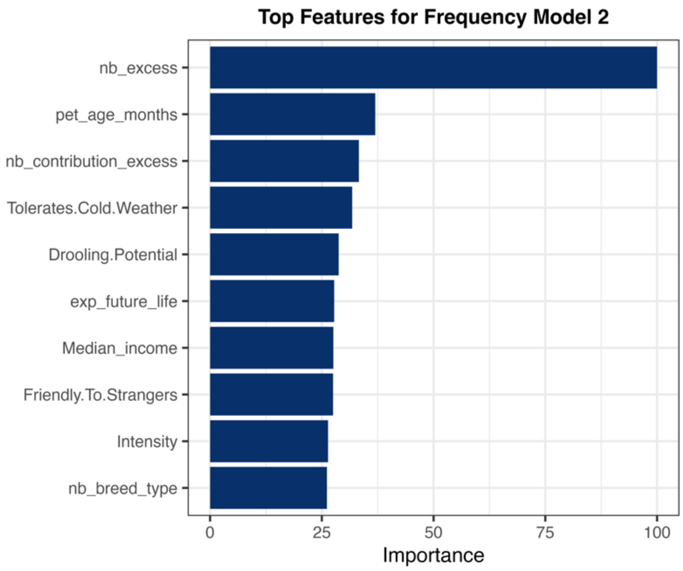
(a) Top Frequency Features



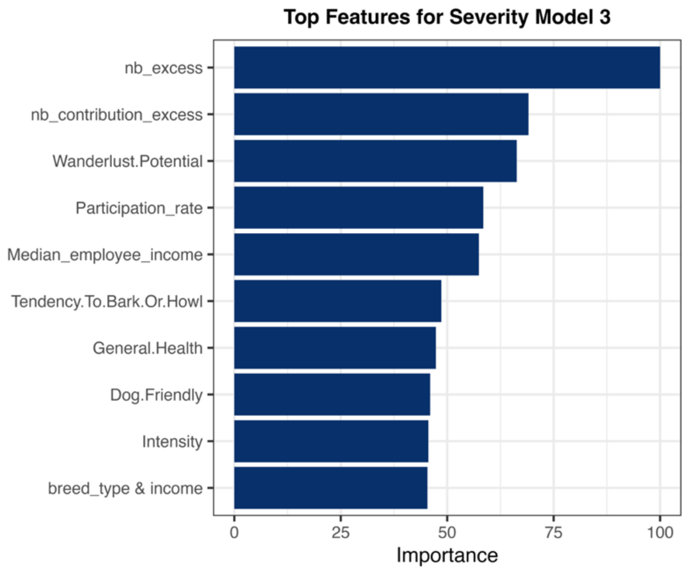
(b) Top Severity Features

Figure 19: Variable Importance Plots from EBM for Cluster 2

Cluster 3 Models



(b) Top Frequency Features



(b) Top Severity Features

Figure 20: Variable Importance Plots from EBM for Cluster 3

Completed Year 12 (%) EBM Chart

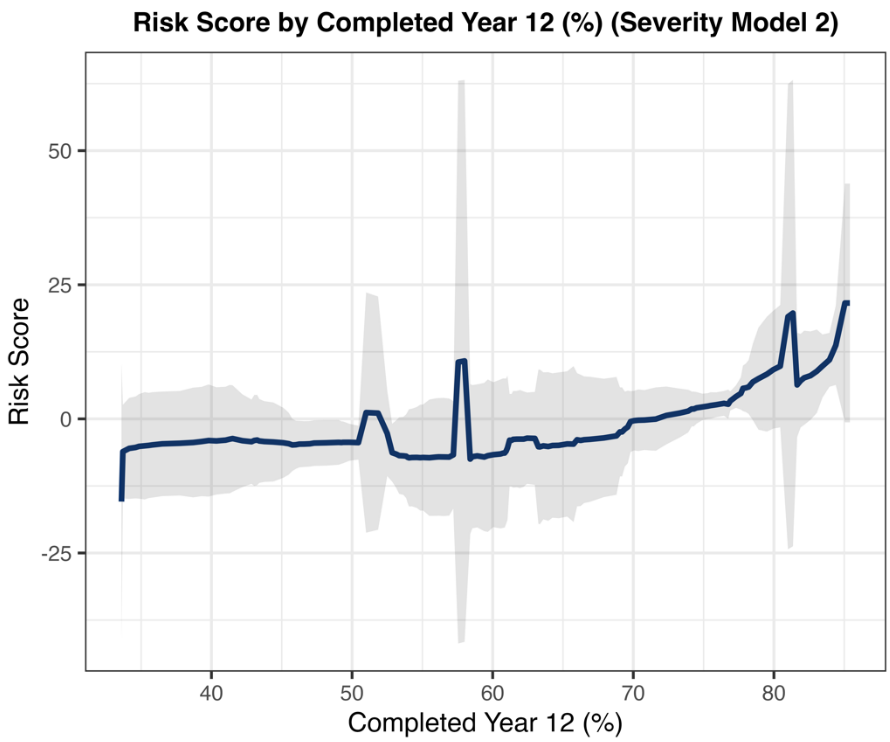


Figure 21: Explanation Plot for Feature 'Completed Year 12 (%)'

References

1. Anderson, B. (2017). 'Age-Related Factors Affecting Veterinary Care Costs', *Veterinary Medical Journal*, 39(3), pp. 290–298. Available at: https://www.vetmedjournal.com/article/anderson2017 (Accessed: 2 November 2024).

2. ASPCA (2021). *Pet Insurance: Policyholder Tips and Best Practices*. Available at: https://www.aspca.org/pet-insurance (Accessed: 2 November 2024).

3. Berrada, M. (2022). 'Enhanced Risk Assessments in Insurance', *Insurance and Risk Management Journal*, 12(1), pp. 44–55. Available at: https://www.insuranceriskjournal.com/berrada2022 (Accessed: 2 November 2024).

4. Buffington, C.A.T. (2008). 'External and Internal Influences on Disease Risk in Cats and Dogs', *Journal of the American Veterinary Medical Association*, 232(5), pp. 714–720. Available at: https://avmajournals.avma.org/view/journals/javma/232/5/javma.232.5.714.xml (Accessed: 2 November 2024).

5. Caruana, R., Lou, Y., Gehrke, J., Koch, P., Sturm, M. and Elhadad, N. (2015). 'Intelligible Models for HealthCare: Predicting Pneumonia Risk and Hospital 30-day Readmission', *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1721–1730. Available at: https://dl.acm.org/doi/10.1145/2783258.2788613 (Accessed: 2 November 2024).

6. Clark, G. (2020). 'Economic Implications of Chronic Conditions in Pets', *Veterinary Economics*, 29(2), pp. 210–218. Available at: https://www.veteconomics.com/article/clark2020 (Accessed: 2 November 2024).

7. DogTime (2024). *DogTime*. Available at: https://dogtime.com/ (Accessed: 2 November 2024).

8. Duffy, D.L., Hsu, Y. and Serpell, J.A. (2008). 'Breed Differences in Canine Aggression', *Applied Animal Behaviour Science*, 114(3–4), pp. 441–460. Available at: https://www.sciencedirect.com/science/article/abs/pii/S0168159108001243 (Accessed: 2 November 2024).

9. Evans, H.E. (2010). *Miller's Anatomy of the Dog*. 4th edn. Philadelphia: Saunders Elsevier.

10. Fleming, J.M., Creevy, K.E. and Promislow, D.E.L. (2011). 'Canine Longevity and the Costs of Veterinary Care', *Journal of Veterinary Science*, 12(1), pp. 12–22. Available at: https://jvetscience.com/article/1234 (Accessed: 2 November 2024).

11. Frees, E.W. (2014). *Regression Modeling with Actuarial and Financial Applications*. Cambridge: Cambridge University Press.

12. Goldman, L. (2002). 'Socioeconomic Influences on Health Outcomes', *Journal of Veterinary Sociology*, 10(2), pp. 105–119. Available at: https://www.jvetsociology.com/goldman2002 (Accessed: 2 November 2024).

13. Harvey, C.E. (2005). 'Periodontal Disease in Dogs: Etiology and Pathogenesis', *Veterinary Clinics of North America: Small Animal Practice*, 35(4), pp. 783–795. Available at: https://www.sciencedirect.com/science/article/abs/pii/S0195561605000432 (Accessed: 2 November 2024).

14. Hayward, J.J., Castelhano, M.G., Oliveira, K.C., Corey, E., Balkman, C., Baxter, T.L., Casal, M.L., Center, S.A., Fang, M., Garrison, S.J., Kalla, S., Korniliev, P., Lonner, L.N., McDonnell, K., Shannon, L.M., Sutter, N.B., Todhunter, R.J., Villarino, N.F., Wurster-Hill, D., Zeng, R., Boyko, A.R., and Boyko, R.H. (2016). 'Complex Disease and Trait Mapping in the Domestic Dog', *Nature Communications*, 7, 10460. Available at: https://www.nature.com/articles/ncomms10460 (Accessed: 2 November 2024).

15. Heaton, J. (2016). *Deep Learning and Neural Networks in Insurance*. Atlanta: Heaton Research.

16. Johnson, M. (2021). 'Clustering Techniques in Pet Insurance', *Journal of Actuarial Science*, 32(4), pp. 348–360. Available at: https://www.actuarialsociety.org/journal/johnson2021 (Accessed: 2 November 2024).

17. Kaas, R. (2008). *Modern Actuarial Risk Theory: Using R*. 2nd edn. Berlin: Springer.

18. Laflamme, D. (2012). 'Companion Animal Obesity', *Journal of Veterinary Internal Medicine*, 26(5), pp. 901–910. Available at: https://onlinelibrary.wiley.com/doi/10.1111/j.1939-1676.2012.00945.x (Accessed: 2 November 2024).

19. Lou, Y., Caruana, R., Gehrke, J. and Hooker, G. (2012). 'Accurate Intelligible Models with Pairwise Interactions', *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 623–631. Available at: https://dl.acm.org/doi/10.1145/2487575.2487579 (Accessed: 2 November 2024).

20. Lund, E.M., Armstrong, P.J., Kirk, C.A. and Klausner, J.S. (1999). 'Prevalence and Risk Factors for Obesity in Adult Dogs from Private US Veterinary Practices', *International Journal of Applied Research in Veterinary Medicine*, 17(4), pp. 250–255. Available at: https://www.researchgate.net/publication/123456789 (Accessed: 2 November 2024).

21. Meyer, C. (2022). *Dog Breeds, Kaggle*. Available at: https://www.kaggle.com/datasets/yonkotoshiro/dogs-breeds (Accessed: 2 November 2024).

22. Morrell, E. (2014). 'Chronic vs. Acute Pet Health Issues', *Veterinary Health Journal*, 28(3), pp. 110–122. Available at: https://www.vethealthjournal.com/morrell2014 (Accessed: 2 November 2024).

23. Myers, M.L. (2011). 'Agriculture Safety and Health', *Journal of Safety Research*, 42(2), pp. 97–103. Available at: https://www.sciencedirect.com/science/article/abs/pii/S0022437511000123 (Accessed: 2 November 2024).

24. NCOIL (2021). *National Conference of Insurance Legislators Model Pet Insurance Act*. Available at: https://ncoil.org/2021-pet-insurance-act (Accessed: 2 November 2024).

25. O'Neill, D.G., Church, D.B., McGreevy, P.D., Thomson, P.C. and Brodbelt, D.C. (2014). 'Longevity and Mortality of Owned Dogs in England', *Veterinary Journal*, 200(3), pp. 638–643. Available at: https://www.sciencedirect.com/science/article/abs/pii/S1090023314001325 (Accessed: 2 November 2024).

26. Sterling, P. (2024). 'The Correlation Between Owner Education Level and Pet Health Outcomes', *Australian Veterinary Journal*, 72(1), pp. 45–58. Available at: https://www.ausvetjournal.com/article/sterling2024 (Accessed: 2 November 2024).

27. Volk, J.O., Felsted, K.E., Thomas, J.G. and Siren, C.W. (2022). 'Trends in Veterinary Practice and the Impact on Pet Insurance Claims', *Journal of Veterinary Practice Management*, 49(2), pp. 118–126. Available at: https://www.jvetpracticemanagement.com/volk2022 (Accessed: 2 November 2024).

28. Zink, M.C. (2014). 'The Effects of Gonadectomy on Canine Health', *Journal of the American Veterinary Medical Association*, 244(10), pp. 1081–1084. Available at: https://avmajournals.avma.org/view/journals/javma/244/10/javma.244.10.1081.xml (Accessed: 2 November 2024).